

```
>> dynare mcmc.mod
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Substitution of exo leads: added 1 auxiliary variables and equations.
Found 23 equation(s).
Evaluating expressions...
Computing static model derivatives (order 2).
Computing static model derivatives w.r.t. parameters (order 2).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
6 block(s) found:
  4 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 15 equation(s)
                                and 15 feedback variable(s).
Computing dynamic model derivatives (order 2).
Computing dynamic model derivatives w.r.t. parameters (order 2).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
7 block(s) found:
  5 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 15 equation(s)
                                and 15 feedback variable(s).

Preprocessing completed.
Preprocessing time: 0h00m00s.
```

STEADY-STATE RESULTS:

q	32.8507
k	4.41224
R	1.01351
b	143.013
x	0.0406367
varphi	0.0303425
mu	0.00686924
kp	2.20776
chi	1.02177
nu	0.00432944
eta	1.28813
zeta	1.02177
N	41.8193
phi	3.41978
Ne	41.5333
Nn	0.286025
xp	6.22046
bp	-101.193
Y	3.45318
eb	0
b_obs	0
Y_obs	0

EIGENVALUES:

Modulus	Real	Imaginary
0	0	0
3.701e-17	3.701e-17	0
3.929e-15	-3.929e-15	0
1.939e-13	1.939e-13	0
3.481e-09	-3.481e-09	0

3.485e-09	3.485e-09	0
2.689e-05	2.689e-05	0
0.9789	0.9789	0
1.018	1.018	0
1.02	1.02	0
1.022	1.022	0
1.026	1.026	0
15.19	-15.19	0
6.334e+18	-6.334e+18	0
4.397e+19	4.397e+19	0
Inf	-Inf	0

There are 8 eigenvalue(s) larger than 1 in modulus for 8 forward-looking variable(s)
The rank condition is verified.

==== Identification Analysis =====

Testing prior mean

The model does not solve for prior_mean (info = 4: Blanchard & Kahn conditions are not satisfied: indeterminacy.)

Try sampling up to 50 parameter sets from the prior.

Found a random draw from the priors that solves the model:

0.0078 0.0105 0.0269 0.0917 -0.1129

Identification now continues for this draw.

Note that differences in the criteria could be due to numerical settings, numerical errors or the method used to find problematic parameter sets.

Settings:

Derivation mode for Jacobians: Analytic using Sylvester equations

Method to find problematic parameters: Nullspace and multicorrelation

coefficients

Normalize Jacobians: Yes

Tolerance level for rank computations: robust

Tolerance level for selecting nonzero columns: 1e-08

Tolerance level for selecting nonzero singular values: 1e-03

REDUCED-FORM:

All parameters are identified in the Jacobian of steady state and reduced-form solution matrices (rank(Tau) is full with tol = robust).

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):

All parameters are identified in the Jacobian of steady state and minimal system (rank(Deltabar) is full with tol = robust).

SPECTRUM (QU AND TKACHENKO, 2012):

!!!WARNING!!!

The rank of Gbar (Jacobian of mean and spectrum) is deficient!

gamma_q is not identified!

MOMENTS (ISKREV, 2010):

All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol = robust).

==== Identification analysis completed =====

You did not declare endogenous variables after the estimation/calib_smoother command.
Initial value of the log posterior (or likelihood): -17452.0463

```
=====
Change in the posterior covariance matrix = 0.0024893.
Change in the posterior mean = 0.1261.
Current mode = -129.4579
Mode improvement = 17581.5043
New value of jscale = 0.025332
=====
```

```
=====
Change in the posterior covariance matrix = 0.0015577.
Change in the posterior mean = 0.092778.
Current mode = -129.6179
Mode improvement = 0.16001
New value of jscale = 0.34439
=====
```

```
=====
Change in the posterior covariance matrix = 0.001419.
Change in the posterior mean = 0.019439.
Current mode = -129.9936
Mode improvement = 0.37566
New value of jscale = 0.74809
=====
```

Optimal value of the scale parameter = 0.74809

Final value of minus the log posterior (or likelihood):-129.993597

MODE CHECK

Fval obtained by the optimization routine: -129.993597

RESULTS FROM POSTERIOR ESTIMATION

parameters

	prior mean	mode	s.d.	prior	pstdev
gamma_q	0.0000	-0.0838	0.0523	norm	0.0500
gamma_N	0.0000	0.0167	0.0220	norm	0.0500
gamma_b	0.0000	-0.0065	0.0263	norm	0.0500

standard deviation of shocks

	prior mean	mode	s.d.	prior	pstdev
eN	0.0100	0.0589	0.0426	inv	0.0050
eY	0.0100	0.2212	0.0213	inv	0.0050

Log data density [Laplace approximation] is 115.072528.

Estimation::mcmc: Multiple chains mode.

Estimation::mcmc: Old mh-files successfully erased!

Estimation::mcmc: Old metropolis.log file successfully erased!

Estimation::mcmc: Creation of a new metropolis.log file.
 Estimation::mcmc: Searching for initial values...
 Estimation::mcmc: Initial values found!

Estimation::mcmc: Write details about the MCMC... Ok!
 Estimation::mcmc: Details about the MCMC are available in mcmc/metropolis\mcmc_mh_history_0.mat

警告: 行列は、特異行列に近いが、正しくスケーリングされていません。結果は不正確になる可能性があります。

RCOND = 1.822623e-16。

> lyapunov_symm (行 153) 内
 lyapunov_solver (行 65) 内
 dsge_likelihood (行 245) 内
 posterior_sampler_iteration (行 64) 内
 posterior_sampler_core (行 199) 内
 posterior_sampler (行 124) 内
 dynare_estimation_1 (行 444) 内
 dynare_estimation (行 105) 内
 mcmc.driver (行 899) 内
 dynare (行 308) 内

Estimation::mcmc: Number of mh files: 1 per block.
 Estimation::mcmc: Total number of generated files: 2.
 Estimation::mcmc: Total number of iterations: 125000.
 Estimation::mcmc: Current acceptance ratio per chain:

Chain 1: 32.7824%
 Chain 2: 33.1448%

Estimation::mcmc: Total number of MH draws per chain: 125000.
 Estimation::mcmc: Total number of generated MH files: 1.
 Estimation::mcmc: I'll use mh-files 1 to 1.
 Estimation::mcmc: In MH-file number 1 I'll start at line 25001.
 Estimation::mcmc: Finally I keep 100000 draws per chain.

MCMC Inefficiency factors per block

Parameter	Block 1	Block 2
SE_eN	90.201	47.940
SE_eY	31.104	29.562
gamma_q	28.943	29.486
gamma_N	46.851	35.103
gamma_b	44.016	34.083

Convergence diagnostics results for chain 1.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 1.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_eN	0.130	0.043	0.000	0.299	
0.282	0.230				
SE_eY	0.225	0.022	0.000	0.191	
0.206	0.243				
gamma_q	-0.038	0.050	0.004	0.521	
0.502	0.478				

gamma_N	0.051	0.022	0.000	0.507
0.457	0.364			
gamma_b	-0.059	0.024	0.000	0.045
0.035	0.027			

Convergence diagnostics results for chain 2.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for chain 2.

p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper
p-val 15% Taper					
SE_eN	0.128	0.038	0.312	0.890	
0.884	0.885				
SE_eY	0.224	0.022	0.000	0.043	
0.037	0.023				
gamma_q	-0.038	0.050	0.000	0.229	
0.237	0.261				
gamma_N	0.051	0.021	0.000	0.235	
0.222	0.197				
gamma_b	-0.059	0.024	0.000	0.241	
0.224	0.227				

Univariate convergence diagnostic, Brooks and Gelman (1998):

Parameter 1... Done!
Parameter 2... Done!
Parameter 3... Done!
Parameter 4... Done!
Parameter 5... Done!

marginal density: I'm computing the posterior mean and covariance... Done!

marginal density: I'm computing the posterior log marginal density (modified harmonic mean)... Done!

ESTIMATION RESULTS

Log data density (Modified Harmonic Mean) is 111.593775.

parameters

	prior mean	post. mean	90% HPD interval		prior	pstdev
gamma_q	0.000	-0.0377	-0.1166	0.0432	norm	0.0500
gamma_N	0.000	0.0513	0.0166	0.0848	norm	0.0500
gamma_b	0.000	-0.0595	-0.0954	-0.0137	norm	0.0500

standard deviation of shocks

	prior mean	post. mean	90% HPD interval		prior	pstdev
eN	0.010	0.1293	0.0695	0.1856	invG	0.0050
eY	0.010	0.2249	0.1908	0.2609	invG	0.0050

Total computing time : 0h40m03s

Note: warning(s) encountered in MATLAB/Octave code

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